

Gene regulatory elements for senolytics:

Category	Example Gene/Element	Function/Role	Relevance to Senolysis / Synthetic Circuit Design
Promoter	p16 ^{INK4a}	Activated in senescent cells	Drives selective expression of senolytic genes in aged or damaged cells.
Repressor	p53	Regulates cell cycle arrest and apoptosis	Acts as a safety checkpoint in synthetic design.
Enhancer	NF-κB enhancer	Activates genes during inflammation and senescence	Amplifies target gene expression in senescent microenvironments.
Terminator	T7 terminator	Ends transcription efficiently	Prevents read-through transcription and maintains modularity.
Sensor/Input	IL-6 / IL-8 (SASP markers)	Secreted cytokines characteristic of senescent cells	Can be used as input signals for senescence detection.
Activator	E2F1	Promotes transcription of genes for cell cycle progression	Helps toggle between senescent and proliferative states.
Reporter	GFP / Luciferase	Produces measurable fluorescent or luminescent signal	Visualizes circuit activation in vitro and in vivo.

Standard Biological Parts from iGEM, Benchling, and SynBioHub:

Part Type	Source Database	Example Part / ID	Function	Potential Use in Senolytic Circuit
Promoter	iGEM Registry	BBa_J23119	Constitutive promoter for baseline expression	Maintains steady control gene output.
Promoter (inducible)	SynBioHub	pBAD (arabinose-inducible)	Activates transcription in presence of arabinose	Enables controllable senolytic activation.
Repressor	Benchling / iGEM	lacI	Inhibits transcription by binding operator	Allows ON/OFF control of therapeutic genes.
Reporter	iGEM	GFP (BBa_E0040)	Produces green fluorescence	Monitor circuit activity visually.
Terminator	SynBioHub	T1 terminator	Stops transcription downstream of a gene	Prevents interference between elements.
Sensor	iGEM / SynBioHub	HSP70 promoter	Responds to heat-shock	Mimics cellular stress linked to senescence.
Effector	Benchling	BAX gene	Pro-apoptotic protein	Drives selective elimination of senescent cells.